On the Credibility of Statistical Inference Within the Neyman-Pearson Hypothesis Testing Framework

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Abstract

Neyman-Pearson based Null Hypothesis Significance Testing belongs to one of the most commonly used methods for testing hypotheses in psychological research. Despite Neyman and Pearson's recommendation that the specification of alpha (α) and beta (β) levels should be adapted given each unique research scenario, clear guidelines for doing so are missing. As a result, the conventional level of $\alpha = .05$ is often used as default without the consideration of β. Moreover, the use of the conventional α-level is argued to be one of the leading causes of the replication crisis. To address this major shortcoming, the goal of the current paper is to propose an intuitive conditional justification of the error rates to guide applied researchers to choose a specific set of alpha and beta for a given research scenario to better reflect the credibility of research findings. More specifically, the selection is based on (i) the proportion of true null hypotheses in the research field and (ii) the conditional probabilities of (in)correct conclusions given the (known) decision to reject or accept the null hypothesis (rather than conditioning on the null and alternative hypotheses being true which is unknown). A Shiny app is available to help researchers with the computation. The method aims to encourage researchers to move away from conventional levels and, instead, strive to rationalize and customize their choices of alpha and beta for each research scenario. This shift aims to facilitate the attainment of credible conclusions, moving beyond automated decision-making based on conventional levels.

Keywords: Alpha, Beta, Null Hypothesis Significance Testing, Conditional Justification

On the Credibility of Statistical Inference Within the Neyman-Pearson Hypothesis Testing Framework

Neyman-Pearson based null hypothesis significance testing (NHST) belongs to the most dominant methods in scientific practice to test a null hypothesis against an alternative hypothesis based on the observed data. Typically, the null hypothesis assumes the absence of an effect, a zero correlation, or no group difference while the alternative assumes that the effect, correlation, or group difference is nonzero (or in a certain direction). The NHST procedure ensures that the researcher's decision "shall not be too often wrong" in the long run (Neyman & Pearson, 1933). This is achieved by the specification of an alpha level (α), which controls the type I error rate of incorrectly rejecting a true null hypothesis, and a beta level (β), which reflects the type II error rate of incorrectly accepting a false null hypothesis. NHST is typically executed by comparing the observed *p*-value (which is always provided using standard statistical software) to a pre-specified alpha level: the null hypothesis is *rejected* if the *p*-value is smaller than alpha and the null hypothesis is *accepted*[1] if the *p*-value is bigger than alpha. This allows researchers to primarily focus on the alpha level, neglecting to consider or justify the beta level and statistical power. This is reflected in research with low prevalence of priori power analysis and prevalence of underpowered studies (Button et al., 2013; McKay et al., 2022). Problematically, conducting research with low power can lead to overestimation of effect sizes and reduced reproducibility of results (Button et al., 2013).

[1] Note that "rejection" and "acceptance" of a hypothesis tell us nothing about the truthfulness of a hypothesis except that the risk of wrongly rejecting or accepting the null is sufficiently low in the long run (Neyman & Pearson, 1933, p. 291). Later, Pearson noted that "some of our wording may have been chosen inadequately" (Pearson, 1955, p. 206; Lehmann, 1993). The exact meaning of the two terms can be seen in the following quote by Neyman in his later writing "It is important, however, to keep their exact meaning in mind and to discard various additional implications which may be suggested by intuition. Thus, to accept a hypothesis *H* means only to decide to take action *A* rather than action *B*. This does not mean that we necessarily believe that the hypothesis *H* is true. Also, if the application of a rule of inductive behavior "reject" *H*, this means only that the rule prescribes action *B* and does not imply that we believe that *H* is false" (Neyman, 1950, p.259–260; Hubbard, 2004). Additionally, describing a result as significant or insignificant is originated from the Fisher's testing theory rather than the one by Neyman-Pearson (Hubbard, 2004). For this reason, the word "significant" is avoided in the present article as it has different meaning based on Fisher's perspective.

The current paper focusses on the question how to specify the alpha and beta levels in NHST to ensure credible statistical inference. From the frequentist's perspective, credible inference is ensured when a decision has a low probability of making the wrong statistical inference given a result of the statistical test[2]. Thus, sufficient credibility would prevent researchers "from making a fool of themselves, and not to make unpublishable results publishable" (Colquhoun, 2014; 1971).

As a motivating example, we consider a group of psychology researchers who are interested in investigating the theory of a priming effect on behavior. The substantive theory of the priming effect asserts that behavior can be triggered automatically on the mere presence of relevant situational features without being mediated by conscious perceptual or judgmental processes (Bargh, et al., 1996). For the current example, researchers hypothesized that "the participants who had been primed with the elderly stereotype would walk more slowly compared to participants who had not been primed with the stereotype-relevant stimuli" (Bargh, et al., 1996, p. 236). Subsequently, the associated null hypothesis assumes that the walking speed of two groups are the same, while the alternative hypothesis asserts that the walking speed of the experimental group is lower than the control group.

To test these statistical hypotheses, the group of researchers needs to formulate the type I and the type II error probability that they are willing to make depending on whether the null or alternative is true. Originally users were encouraged to specify these error rates for each unique research scenario and each unique testing problem (Neyman & Pearson, 1933). In current statistical practice, however, researchers often rely on the default (but arbitrary) choice of an alpha level of 0.05 and omit a power analysis (McKay et al., 2022), and thus the actual type II error rate is generally unknown. The default choice of 0.05 for alpha may be

[2] The given frequentist definition is inspired by Ioannidis (2005).

attributed to Sir Ronald Fisher's initial suggestion (Fisher, 1958, p.16; Bangdiwala, 2016)[3], even though Fisher later argued the use of a default across all research scenarios for making an automatic conclusion is "absurdly academic" (Fisher, 1973, p. 44-45; Lehmann, 1993)[4]. Despite these arguments by the statisticians who developed the statistical theory, the researchers tended to stick to conventional error probabilities to test the theory of priming in their research design.

The neglect of specifying and justifying the error rates for each unique testing scenario may be attributed to the fact that practical and concrete guidelines for determining alpha and beta levels are largely absent. This leads testers to (arbitrary) conventional levels in accordance with "standard statistical practice" in accordance with Sir Ronald Fisher's initial notes. Moreover, the use of conventional levels may give researchers the illusion of objectivity (Berger & Berry, 1988). In fact, however, it has been argued that the use of such conventional error rates (such as $\alpha = .05$) may be one of the causes of the replication crisis because "statistical standards of evidence for claiming new discoveries in many fields of science are simply too low" (Benjamin et al., 2017, p.6). This suggests that the current application of the testing procedure in fact did not protect researchers from being wrong too often in the long run, as originally intended by Neyman and Pearson (1933). For this reason, there has been a call to researchers to justify the error rates that are used when performing NHST (Lakens, et al., 2018). Benjamin et al. (2017) proposed lowering the default to $\alpha = .005$, while Maier and Lakens (2022) proposed a compromise power analysis for specifying alpha and beta.

[3] "The value for which P = 0.05, or 1 in 20, is 1.96 or nearly 2; it is convenient to take this point as a limit in judging whether a deviation ought to be considered significant or not" (Fisher, 1958, p.16; Bangdiwala, 2016).
[4] "A man who "rejects" a hypothesis provisionally, as a matter of habitual practice, when the significance is 1% or higher, will certainly be mistaken in not more than 1% of such decision… However, the calculation is absurdly academic, for in fact no scientific worker has a fixed level of significance at which from year to year, and in all circumstances, he rejects hypotheses; he rather gives his mind to each particular case in the light of his evidence and his ideas" (Fisher,1973, p. 44-45; Lehmann, 1993).

Now let us come back to the example where the researchers are interested in investigating the applicability of primming effect in a specific scenario. In practice, substantive theories may play a role in certain situations but not in other situations. This would suggest that in certain experimental designs priming would influence behavior (i.e., H_1 is true) while in other designs priming would not influence behavior (i.e., H_0 is true). To visualize this, we can think of a population of hypotheses where there is an unknown proportion p_1 of experimental designs in which H_1 is true (i.e., where the theory is applicable and priming does have an effect) and a remaining proportion p_0 where H_0 is true (i.e., where the theory is not applicable and priming has no effect). Figure 1 displays this population of true hypotheses as a big cloud with a certain proportion of true nulls and the remaining proportion of true alternatives. For the current experimental design (i.e., the effect of priming on walking speed), the true (but unknown) hypothesis $H_?$ can be seen as a "draw" from this hypothetical population of true hypotheses. Thus, in order to understand the credibility of a decision when rejecting or accepting the null hypothesis at hand, it seems imperative to take the proportions p_0 and p_1 into account from which the current hypotheses are "drawn". Similarly, Ioannidis (2005) and Benjamin et al, (2017) consider the proportion of true null and alternative hypotheses within a research field when evaluating the credibility of research findings. Colquhoun (2014) also emphasizes the importance of considering the proportion of true nulls to ensure credible statistical inference. In statistical practice, however, these proportions are generally ignored. This can be explained by the fact that the *classical error probabilities* condition on the null and alternative being true, i.e., α = P(wrong decision | H_0 is true) and β = P(wrong decision | H_1 is true), allowing researchers to unduly ignore these proportions.

[Figure 1 about here]

A related issue of classical error rates in NHST is that they do not provide researchers with an intuitive quantification of the credibility of their statistical decisions. As noted, classical error probabilities quantify the rate of wrong decisions given the null or alternative being true, and thus, these error rates condition on something that is in fact unknown, yet they are assumed to be true. Instead, the credibility of statistical decisions would be clearer if the researchers focus on the rate of wrong decisions by conditioning on the known decisions. Thus, in abstract terms, rather than emphasizing on α = P(wrong decision | H_0 is true) and β = P(wrong decision | H_1 is true) for a given test, it is more informative to consider P(wrong decision | reject H_0) and P(wrong decision | accept H_0). We shall denote these alternative conditional error probabilities as γ and κ, respectively. If these conditional error probabilities are known, we would know the average rate of incorrect conclusions in the case of a rejection of the null and the average rate of incorrect conclusions in the case of an acceptance of the null. Thus, instead of focusing on the error rates conditional on the unknown truth (Figure 1), we argue to focus on the conditional error rates given the known decisions. Thereby researchers have a clearer notion about credibility of their statistical decisions. As will be shown in this paper, the proportions p_0 and p_1 introduced above are necessary to obtain these alternative conditional error rates.

The present article builds on these insights and further extends the idea of conditional probabilities in NHST by introducing a procedural framework to implement these recommendations in statistical practice to give a better reflection of the credibility of research findings. We propose a *conditional justification* for error rate selection that can guide researchers to select a specific combination of alpha and beta in their unique research scenario to reach a sufficiently credible conclusion. The credibility of a conclusion pertains to the degree of trustworthiness of a statistical testing decision when executing a NHST by *conditioning on the decision* rather than *conditioning on the true hypothesis* which is

unknown (as is the key focus in NHST). More specifically, a credible statistical inference has a sufficiently low long-term probability of making an incorrect inference based on the results of NHST. By employing the presented method, (i) researchers get a better understanding of the credibility of their conclusions and (ii) researchers are better able to justify and deliberate their choices of alpha and beta for a given research scenario, thus avoiding the mechanical decision-making commonly tied with conventional levels.

Motivation for the Conditional Justification

In essence, the proposed conditional justification seeks to address two shortcomings inherent in Neyman-Pearson based NHST. Firstly, the conditional justification prioritizes the attainment of a credible conclusion by focusing on the '*conditional error rates*' which are conditional on the decision of the test (i.e., to accept or reject the null) rather than the '*classical error rates*' which condition on the null or alternative being true. Secondly, the presented method advocates the specification of alpha and beta levels for each unique research scenario by taking the proportion of true null hypotheses in the field of research or context at hand into account. This proportion is ignored in NHST as originally proposed by Neyman and Pearson.

As displayed in Table 1, the classical error rates α and β, which are also referred to as the False Positive Rate (FPR) and False Negative Rate (FNR), condition on the null and alternative hypotheses being true. Rather than conditioning over the rows as in Table 1, we propose to condition over the columns to obtain the error rates of incorrect conclusions conditional on the decisions when accepting or rejecting the null, as displayed in Table 2.

[Table 1 and 2 about here]

The conditional error rates of incorrect conclusions given a statistical decision (Table 2) are referred to as the False Discovery Rate (FDR) and False Omission Rate (FOR), which we shall denote by the Greek symbols gamma (γ) and kappa (κ), respectively. These

quantities directly follow from the application of Bayes' theorem. Here, γ is defined by the ratio of false positives (i.e., incorrect rejections of the null) relative to the total number of positives (i.e., rejections). In other words and in the context of NHST, γ represents the proportion of true nulls among all rejections of the null. Thus, γ quantifies the average rate of false conclusions in the case of a rejection of the null. Mathematically, we can write:

$$\gamma = p(H_0 \; is \; true | rejecting \; H_0) = \frac{p(H_0 \; is \; true \; \& \; rejecting \; H_0)}{p(rejecting \; H_0)} = \frac{p_0 \alpha}{p_1(1-\beta) + p_0 \alpha}, \; (1)$$

where the type I and the type II error rates are denoted as the usual α and β, and p_1 and p_0 reflect the proportions of true alternative hypotheses and true null hypotheses in a population of hypotheses from which the current true hypothesis is drawn, respectively, with $p_1 + p_0 = 1$. These proportions will be discussed later. Similarly, κ is defined as the ratio of false negatives (i.e., incorrect acceptance of the null) relative to the total number of negatives (i.e., acceptance). In the context of NHST, κ is the proportion of true alternatives among all cases where the null is accepted. Thus, κ quantifies the average rate of false conclusions in case of acceptance of the null. Mathematically, we can write

$$\kappa = p(H_0 \; is \; false | accepting \; H_0) = \frac{p(H_0 \; is \; false \; \& \; accepting \; H_0)}{p(accepting \; H_0)} = \frac{p_1 \beta}{p_0(1-\alpha) + p_1 \beta} \; (2)$$

Because these alternative error rates condition on the known statistical decision after observing the data, the credibility of the two possible decisions are directly clear for a given test (Figure 2).

[Figure 2 about here]

To come back to the introductory example, let us assume that the proportion of true nulls in the field of priming research is equal $p_0 = p_1 = 0.5$. One way to interpret this proportion is to think of an infinitely large population of research scenarios and the theory of priming is applicable in 50% of these scenarios and/or experimental designs. Thus, this population consists of 50% true null hypotheses (where priming is not applicable) and 50%

true alternatives (where priming is applicable). The current hypothesis regarding the applicability of priming in the current research scenario (i.e., the effect of priming on walking speed) can be seen as a 'draw' from this population; see also Figure 1. Moreover, let us consider a hypothesis test with classical error rates of α = .05 and β = .2. The conditional error rate of true nulls among all rejections is then equal to

$$\gamma = \frac{.5 \times .05}{.5(1 - .2) + .5 \times .05} = .059.$$

Hence, the probability of an incorrect decision in the case of a rejection of the null is around 6%. The conditional error rate of true alternatives in case of acceptance of the null is equal to

$$\kappa = \frac{.5 \times .2}{.5 \times (1 - .05) + .5 \times .2} = .174.$$

Hence, the probability of an incorrect decision in case of accepting the null is around 17%. Interestingly, the conditional error rates gamma and kappa are quite close to the classical error rates alpha and beta, respectively. This is a consequence of the choice of equal proportions of true nulls and true alternatives.

The importance of the proportions of true nulls and true alternatives in a hypothetical population can be demonstrated by conclusions from the Open Science Collaboration (2015) where it was shown that the probability of replicating positive (non-zero) findings differ between different sub-fields within psychology. This suggests that the proportion of true nulls and true alternatives can differ across research fields and sub-fields. For this reason, different choices for p_0 may be preferred in different research scenarios rather than using the same proportion regardless of the context. For example, using the results of the Reproducibility Project: Psychology (Open Science Collaboration, 2015), Johnson et al. (2017) showed that the proportion of true null hypotheses in the broad field of psychological science may be as large as .93 when taking publication bias into account, which thus implies a proportion of true alternatives of only .07. For this reason, for illustrative purposes let us use the

proportions $p_0 = .93$ and $p_1 = .07$ (instead of $p_0 = p_1 = .5$) to see how this would affect the probabilities of drawing the wrong conclusion when conditioning on the statistical decisions, while still using the same standard classical error rates of $\alpha = .05$ and $\beta = .20$. In this case, the conditional error rates would be

$$\gamma = \frac{.93 \times .05}{.07 \times (1 - .2) + .93 \times .05} = .454$$

$$\kappa = \frac{.07 \times .2}{.93 \times (1 - .05) + .07 \times .2} = .016.$$

Thus, the proportion of drawing the wrong conclusion is approximately 45% when rejecting the null hypothesis. In other words, the probability of drawing the correct conclusion when rejecting the null would be almost identical to flipping a fair coin. Thus, if the proportion of true nulls may be as large as .93 as argued by Johnson et al. (2017), the credibility of research findings is seriously at stake when using conventional classical error rates.

Figure 3 further demonstrates how the conditional error rates of false conclusions γ given a rejection (solid line) and κ given an acceptance (dashed line) depend on the proportion of true nulls when using the conventional levels α = .05 and β = .2. An interactive version of Figure 3 is also available in https://tkwong3004.shinyapps.io/ErrorRates/. The credibility of positive findings diminishes as the proportion of true null hypotheses increases, leading to a higher value of conditional error rate γ, as in the example where γ is increased from .059 to .454. In contrast, the credibility of null findings increase with the rises in p_0 resulting in a smaller value of the conditional error rate κ. In the example, the value of κ is decreased from .174 to .016. Moreover, it is reasonable to suspect that the conditional error rate γ may be even higher than 45.4% because of the prevalence of questionable research practices (John, Loewenstein, & Prelec, 2012) and underpowered studies with small sample sizes (Bakker, Van Dijk, & Wicherts, 2012; Button et al., 2013; Nuijten, et al., 2020), among other factors.

[Figure 3 about here]

It should be noted that the conditional justification implicitly assesses the evidentiary impact of a testing decision by considering the comparative plausibility of testing outcomes under the two competing hypotheses (Bayarri et al., 2016). This nuanced difference contrasts with classical NHST, which primarily centers on the potential for classical errors when either one of the hypotheses is true. In fact, the odds of a correct conclusion given a rejection and acceptance based on Equation (1) and (2) can be written as:

$$\frac{1-\gamma}{\gamma} = F_{rejection} \times \frac{p_1}{p_0} \tag{3}$$

$$\frac{1-\kappa}{\kappa} = F_{acceptance} \times \frac{p_0}{p_1} \tag{4}$$

where $F_{rejection} = \frac{1-\beta}{\alpha}$ and $F_{acceptance} = \frac{1-\alpha}{\beta}$. The changing factors, $F_{rejection}$ and $F_{acceptance}$, indicate the relative plausibility of the testing result given the two hypotheses and quantify the evidentiary impact of the testing decision for the null hypothesis relative to the alternative hypothesis. Thus, given these relations, one can derive appropriate classical error rates (α and β) to arrive at a sufficiently credible conclusion by appropriately specifying the target values of γ and κ as well as the proportion of true null hypothesis in the field of research with careful deliberation. We come back to this later in this paper.

Lastly, it is important to note that despite the use of Bayes' rule to derive the conditional error rates, our approach remains a frequentist procedure. Of course, the proportion of true nulls and the proportion of drawing the wrong conclusion when rejecting the null hypothesis also have a clear Bayesian interpretation as the prior probability of the null being true and the posterior probability of the null being true given a rejection of the null. However, the Bayesian interpretation of probability as subjective degree of belief (or

credence) is unnecessary here as the proportion of true nulls and the conditional probabilities are long-term probabilities within frequentist's school of thought (see also, Colquhoun, 2014, for a related discussion). Relatedly, Colquhoun (2019) also recommended the use of conditional error rates given the outcome of the test. Unlike our suggestion however he used the observed *p*-value (or likelihood ratio) as input, which can only be done *after* observing the data. Our suggestion is to use the conditional error rates (γ and κ) to check the credibility of the statistical test and to determine the classical error rates (α and β) accordingly *before* collecting the data. Next we elaborate how to obtain the classical error rates (which are required for NHST) using the alternative conditional error rates into account as well as the proportions of true nulls and true alternatives in a given research scenario.

The Procedure of Conditional Justification

The choice of the classical error rates incorporating the proposed conditional justification hinges on specifying the desired level of credibility of the decision (as indicated by the associated conditional error rates γ and κ) and the proportion of true null and alternative hypotheses in a population of hypotheses (indicated by p_0 and p_1). This can be done using the following steps:

1) *Specify the proportions p_0 and p_1.* For the given field or research in which the test is conducted, which can be relatively broad (e.g., 'social psychology', 'neuropsychology', or even 'psychology' as a whole) or relatively specific (e.g., the research on "priming" or "Type D personality"), one needs to specify the proportion p_0 of true null hypotheses (where an effect, correlation, or group difference is zero[5]) and the proportion p_1 of true alternative hypotheses (where an effect, correlation, or group difference is nonzero).

[5] Possibly the proportion of true null hypotheses p_0 may also contain true negligible effects.

2) *Choose the conditional error rates γ and κ.* For the hypothesis test at hand, one needs to specify the desired levels of the conditional error rates, γ (in case of rejection) and κ (in case of acceptance), depending on the desired credibility of the statistical conclusion.
3) *Derive the classical error rates α and β.* Plug the proportions p_0 and p_1 and the desired γ and κ levels in the following two equations to obtain the classical error rates α and β:

$$\alpha = \frac{\gamma(1-p_0)(p_0-1+\kappa)}{p_0(p_0-\kappa p_0-\gamma p_0+\kappa+\gamma-1)} \quad (5)$$

$$\beta = \frac{\kappa(\gamma-p_0)}{p_0(1-\kappa-\gamma)+\kappa+\gamma-1} \quad (6)$$

The application of the procedure is briefly shown via the introductory example on priming research and the guidelines for the specification with justification are provided in later section:

1) The proportions p_0 and p_1 are specified to be .5 since at the time Bargh et al., (1996) conducted their study, the effect of priming on behavioral response had not yet to be investigated. For this reason, one may use the Principle of Insufficient Reason (Jeffreys, 1961) to argue for equal proportions p_0= p_1=.5.
2) The desired levels of the conditional error rates, γ and κ are specified to be .01 for illustrative purposes. This implies that in the long run only 1% of all research findings result in false conclusions.
3) The classical error rates should be the following for ensuring a credible conclusion based on NHST

$$\alpha = \frac{.01(1-.5)(.5-1+.01)}{.5(.5-.01\times.5-.01\times.5+.01+.01-1)} = .01.$$

$$\beta = \frac{.01(.01-.5)}{.5(1-.01-.01)+.01+.01-1} = .01$$

Although a conventional alpha level of .01 is recommended, the suggested beta level is substantially smaller than the conventional .2.The mathematical derivation for Equation (5) and (6) are provided in Appendix A. A Shiny App (https://tkwong3004.shinyapps.io/ErrorRates/) is also available for the selection of the classical error rates with the source code being available at https://osf.io/6nuq5/. The R codes for generating the figures and a brief tutorial on the R function for the selection of error rates are also available in the same OSF repository.

Illustrative Applications

The following examples highlight the importance of basing the classical error rates α and β on the conditional error rates γ and κ, and the proportion of the true nulls. As will be motivated, ignoring these components and solely relying on classical error rates result in statistical decisions with limited credibility of the conclusions. The reader can use the Shiny web app or the R tutorial on OSF to follow the empirical applications for conducting conditional justification. Guidelines for the specification of proportion of true nulls and conditional error rates are provided in the subsequent section.

Empirical Application I

Xia et al. (2023) investigated the influence of music environment on conceptual design creativity and hypothesized that "both pure music (*Authors' note: classical music*) and intelligible semantic music background (*Authors' note: poems*) have a positive effect on divergent thinking" (p. 3). Divergent thinking refers to the "process of generating creative ideas by proposing multiple solutions" (p.1). A total of 60 participants were recruited and randomly assigned to one of three conditions (pure music background, intelligible semantic music background and quiet background). An ANOVA without power analysis was

conducted to test if there is a difference in divergent thinking between the groups. The null hypothesis was accepted[1], $F(2, 54) = 0.74$, $p = 0.481$ (Xia et al., 2023).

Conducting NHST without a power analysis and elaboration on the selection of classical error rates can lead to uninformative results with conclusions lacking credibility. Suppose that the effect size was $f = .25$ (medium) with $\alpha = .05$, the power of the test would be .37. Given the proportion of true null hypothesis in the field of psychology is estimated to be as large as .93 accounted for possible publication bias (Johnson, et al., 2017), the resulting conditional error rates are in fact equal to $\gamma = .642$ and $\kappa = .048$ based on Equation (1) and (2). Hence we see that in the case of a rejection of the null hypothesis, the null hypothesis would still remain more probable than the alternative hypothesis, as the probability of a false conclusion given a rejection is .642. This example underscores the importance of not only conducting power analysis but also considering the proportion of true null hypothesis and their implications for the credibility of the conclusions drawn from NHST. Note here that since the null hypothesis is accepted, the probability of a false conclusion is relatively small (only around .05 on average).

Suppose a conditional justification is considered with a proportion of true null hypotheses of .93 (to account for possible publication bias), and we desire to change the initial odds of true nulls relative to the true alternatives based on the outcome of the test with a factor of at least 19, then it can be shown (as elaborated in the next section) that the conditional error rates will be $\gamma = \kappa = .004$. The use of the Shiny app for the calculation of the suggested classical error rates is shown on the top side of Figure 4. Under these conditions, the required classical error rates for arriving at a credible conclusion from the test should be $\alpha = .00028$ and $\beta = .05262$, which are considerably smaller than the conventional error rates.

[Figure 4 about here]

Empirical Application II

Bem (2011) investigated the existence of precognition and premonition (i.e., the ability to see the future) in various experimental settings and concluded that "The presentiment studies provide evidence that our physiology can anticipate unpredictable erotic or negative stimuli before they occur. Such anticipation would be evolutionarily advantageous for reproduction and survival if the organism could act instrumentally to approach erotic stimuli and avoid negative stimuli" (Bem, 2011, p. 408). A one-tailed one-sample *t* test was conducted assuming an effect size of $d = .25$ under the alternative hypothesis with $\alpha = .05$, $\beta = .2$, and $N = 100$. The null hypothesis was rejected[1], $t(99) = 2.51$, $p = .01$ (Bem, 2011).

As opposed to the previous study did not consider a power analysis, Bem (2011) did conduct a power analysis to ensure that the test was sufficiently powerful. Nevertheless, the neglection of the proportion of true null hypothesis in the field of pre-cognition in combination with conventional error rates is prone to unreliable conclusions. Presumably, the neglection of this specification implicitly implies that the hypothesis at hand comes from a population with a proportion of true null hypotheses of $p_0 = 0.5$ implying that in half of the population of hypotheses premonition actually exists. The resulting conditional error rates are $\gamma = .059$ and $\kappa = .174$. As the null is rejected, the conditional error of a false conclusion would be around 6% on average. Even though this conditional error rate seems small, this is a direct consequence of choosing p_0 equal to 0.5. This choice however may conflict with the general viewpoint about premonition in scientific practice. Given the lack of established physical and biological theories, as well as empirical evidence supporting the notion of future events retroactively influence people's behavior (Wagenmakers et al., 2011), setting equal

proportions in favor and against the existence of premonition would not correspond with the general scientific point of view. Instead, a more pessimistic choice about the alternative would better fit this viewpoint. If we would set the proportion of true null hypotheses equal to .99, and thus the proportion of true alternative hypotheses where premonition actually exists to be to .01, the conditional error rates would be $\gamma = .861$ and $\kappa = .002$, indicating a chance of incorrect conclusion in the case of rejection of approximately 86%. Therefore, when also taking into account the proportion of true null hypotheses based on the general viewpoint in science regarding the existence of premonition and subsequently report the corresponding conditional error rates, one obtains a better understanding of the credibility of this scientific finding. Finally note that if the desired conditional error rates would be set to $\gamma = \kappa = .0005$ (see later section for the justification of the selection), the classical error rates should be set to $\alpha = .000005$ and $\beta = .05263$ (see Figure 4 for the calculation).

In summary, both examples underscore the importance of selecting appropriate classical error rates (α and β) in relation to the proportion of true null hypotheses and the credibility of the conclusions as indicated by the error probabilities conditional on the decision. More specifically, the first application illustrates a scenario in which the statistical test's evidentiary impact is so minimal that the null hypotheses remains more probable to be true than the alternative, regardless of the test result. The second application emphasizes the required strength of evidence indicated by test for arriving at credible conclusion depends on the initial proportion of true null hypotheses in a population that reflects the general scientific viewpoint about a certain theory.

Guidelines for the Specification of the Proportion of True Null Hypotheses and Conditional Error Rates

This section offers guidelines for selecting the proportion of true null hypotheses and the conditional error rates. Once the classical error rates have been specified using the conditional justification, the sample size still needs to be determined for the hypothesized effect size. For guidance on how to hypothesize effect sizes, readers are referred to Durlak (2009) and Lakens (2022), for example. The summary of the guidelines and the procedure of conditional justification are provided in Table 3.

[Table 3 about here]

Selecting the proportion of true null hypotheses

The choice of the proportion of true null hypotheses in the (exchangeable) population of true hypotheses plays an important role in the proposed conditional justification of the error rates in Neyman-Pearson based NHST. These proportions need to be explicitly motivated and stated in order to obtain the conditional error rates gamma and kappa (unlike the classical error rates alpha and beta which can be specified without taking the proportions into account). Thus, the conditional justification require more deliberation and transparency when determining the error rates and reflecting the credibility of research findings in comparison to classical Neyman-Pearson based NHST. Moreover, it is possible that other researchers in the community can re-do an analysis using other values for these proportions to assess the credibility of the conclusions from a different viewpoint. We distinguish two key factors that affect the choice of these proportions. The first is the broadness of the research field and the second is the general viewpoint of the target audience.

(1) Broadness of the research field - To determine the proportion of true nulls in the population from which the current hypothesis is "drawn", a key question is how broad the field of research under consideration is. We distinguish three levels of broadness of the research field.

- **Very broad**. An entire field of research is considered, such as, psychology, sociology, or medical research. For example in psychology, the proportion of true nulls can be then based on a population of true null and alternative hypotheses in the entire field of psychological research.
- **Moderately broad**. A subfield of research is considered, such as, social psychology, developmental psychology, or clinical psychology, all within the broader field of psychology.
- **Very specific**. A specific research topic or theory is considered, such as the theory on the presence of the priming effect, the theory of cognitive dissonance, or the theory of bystander effect (within the subfield of social psychology). Then, in the introductory example, the proportion of true nulls may be based on the proportion of research scenarios where the theory of priming does not have an effect.

Depending on the available information and the viewpoint of the research team, a specific level of broadness may be preferred. For instance, when conducting large-scale reproducibility projects concerning whole research fields (e.g., Reproducibility Project Psychology, or Reproducibility Project: Cancer Biology), it is preferred to specify the proportion of true nulls in relation to the entire respective field. On the other hand regarding the applicability of a single substantive theory in a specific experimental study (as in the introductory example) or in a specific multilab investigation (such as the multilab replication of the theory of cognitive dissonance by Vaidis et al., 2024), specifying the proportion in relation to the specific theory is advisable. In general, it would be best that the considered field of research is as specific as possible in order to have the most accurate reflection of the hypotheses at hand in relation to the (hypothesized) population of true null and alternative hypotheses.

To specify the proportion, different approaches can be considered. Johnson et al. (2017) estimated that the proportion of true nulls in psychological science may be as large as .93 when taking publication bias into account based on the results of the Reproducibility Project: Psychology (Open Science Collaboration, 2015). One can also base the proportions using the approximate reproducibility rates of known research field (Table 4). Alternatively, a meta-analysis could be considered to elicit the proportion of the true nulls for the (sub)field of research at hand (e.g., Bartoš et al., 2024). In the assessment, it is crucial to take the possibility of questionable research practices and the magnitude of the file-drawer problem. Lastly, scientific experts could be consulted to determine a realistic proportion of true nulls in the field of research at hand. This could be achieved for instance using questionnaires. More specifically, the Delphi method can be used for reaching a general opinion consensus of a group of experts (Dalkey & Helmer, 1963).

[Table 4 about here]

(2) Standpoint of the audience - The choice of the proportions can also depend on the standpoint of the target audience. For instance, when the goal is to convince a pessimistic audience about the presence of an effect, it may be preferred to specify the proportion of true null hypotheses to be relatively large to reflect a pessimistic viewpoint. The underlying reasoning is that a large proportion of true null hypotheses require greater evidentiary impact from the statistical test for reaching a credible conclusion, in other words, “extraordinary claims require extraordinary evidence” (Wagenmakers, et al., 2011, p. 428). Otherwise, the credibility of a research finding would be overestimated due to an optimistic selection of the proportion of true null hypotheses which conflicts with the general viewpoint. We distinguish the following three general categories.

- **Pessimistic.** When the aim is to convince a pessimistic audience about the truthfulness of the alternative (or null) hypothesis, the proportion of true nulls (or the proportion of true alternatives) should be relatively large. In Empirical Application II, within the specific research domain investigating the existence of premonitions, the target audience (i.e., the majority of the scientific community) holds a pessimistic perspective regarding its existence. Therefore, a pessimistic approach should be adopted.
- **Neutral.** When the target audience has a neutral standpoint about the truthfulness of the hypotheses, the proportion of true nulls could be moderately sized. This setting may be applicable when considering a completely new (sub)field of research or new research topic. It may then be justifiable to follow the Principle of Insufficient Reason (Jeffreys, 1961) resulting in equal proportions $p_0 = p_1 = .5$. The introductory example would provide a good illustration of applying a neutral choice of proportion.
- **Optimistic.** When the target audience is relatively certain about the truthfulness of the alternative (or null) hypothesis, the proportion of true nulls (or the proportion of true alternatives) can be relatively small. This setting may be applicable in replication studies where most studies concluded that a theory is applicable and the aim of the replication study is to confirm this with new evidence.

Specifying the conditional error rates γ and κ

The conditional error rates gamma and kappa give a direct quantification of the average credibility of research findings by conditioning on the decision. Thus, it is crucial to have guidelines to aid researchers in the specification of γ and κ. We can distinguish the following guidelines and requirements:

(G1). A minimal requirement is that $\gamma < \mathbf{0.5}$ ***and*** $\boldsymbol{\kappa} < \mathbf{0.5}$, so that the chance of drawing the wrong conclusion on average is less than the chance of heads in a fair coin toss. More importantly, the violation of this requirement leads to contradictory conclusions, such as in Empirical Application I, in which the null would on average be more likely than the alternative despite the rejection of the null.

(G2). If the credibility of positive findings and null findings are deemed equally important, equal conditional error rates should then be chosen: $\gamma = \kappa$.

(G3). In order for a study to sufficiently inform the scientific community about the truthfulness of a hypothesis, it can be argued that the conditional error rates (γ and κ) should be smaller than the initial proportions by a certain minimal 'changing factor', say, $F_{rejection}$ in the case of rejection and $F_{acceptance}$ in the case of accepting the null (see also Bayarri et al., 2016, for a related discussion). This guideline ensures that we learn enough from the research given the financial investments and the researchers' efforts, and to ensure that the outcomes of research has a sufficient impact on the trajectory of future studies.

The concept of the changing factor (G3) was used to specify classical error rates to obtain sufficiently low conditional error rates. Depending on the decision (i.e., to reject H_0 or not), the changing factors quantify the change of the initial odds of proportion of true nulls to true alternatives to the odds of the conditional error probabilities according to Equation (3) and (4). For instance, classical error rates of $\alpha = \beta = .05$ would imply that γ and κ should be lower than p_0 and p_1 by a changing factor of 19 when $p_0 = .5$. In order to fulfill (G1), (G2) and (G3) simultaneously, the conditional error rates should be:

$$\gamma = \kappa = \frac{1}{19 \times \frac{p_1}{p_0} + 1} \quad \text{when } p_1 \geq .5$$

$$\gamma = \kappa = \frac{1}{19 \times \frac{p_0}{p_1} + 1} \text{ when } p_0 > .5.$$

The selection of conditional error rates in Empirical Application I and II made use of the two equations above. To further increase the credibility of the decisions, one can opt for a more substantial changing factor. The value of this depends on the researcher's tolerance for the cost of false conclusions. In general, a larger value implies a more credible conclusion. The resulting error rates using different changing factors with different proportions of true nulls are shown in Table 5. The changing factor of 19, 99 and 199 are displayed as these value correspond to α of .05, .01 and .005 when p_0 = .5. It is important to emphasize that these conventional α levels should be applied in conjunction with the appropriate β levels to ensure the attainment of a credible conclusion. Moreover, these conventional thresholds are adjusted and tailored to each distinct research scenario, taking into account the proportion of true null hypotheses.

[Table 5 about here]

The impact of this changing factor on the suggested error rates is illustrated in Figure 5, revealing that as the proportion of true nulls increase, α decreases while β increases. Increasing the changing factor leads to a decrease of the classical error rates in general. Figure 6 illustrates the influence of the proportion of true nulls on the required sample size for a given effect size. It becomes apparent that as the proportion of true nulls deviates from .5, a larger sample size is needed to achieve the specified γ and κ. This is because when a hypothesis reflects a theory that often has been shown to be true, more data are required to sufficiently “challenge” this hypothesis and to obtain a sufficient level of credibility for the opposing hypothesis.

[Figure 5 about here]

[Figure 6 about here]

Discussion

Neyman-Pearson based NHST belongs to the most dominant methods to test statistical hypotheses in applied statistical practice. In current statistical practice, this testing framework is mixed with Fisherian testing where a *p*-value is simply compared to a prespecified type I error rate (typically .05), from which it is decided to reject the null or not, while reporting an idealized type II error rate (but which actually is often ignored). Despite their popularity, we argue that the methodology is of limited use in ensuring the credibility of research findings because of the following two reasons. First, the classical error rates condition on the null and alternative hypothesis to be true while in fact their true state is unknown (which is the reason for testing the hypotheses in the first place!). We argue that this counterintuitive conditioning may be an important reason for researchers to refrain from carefully specifying the classical error rates under the given research scenario (as also recommended by Neyman & Pearson) but rather to follow conventional error rates. As argued, this behavior, which has become the standard in statistical practice, may be an important reason for the reproducibility crisis (Benjamin et al., 2017). The crisis has shown us that the current implementation of the testing procedure does *not* protect researchers from drawing the wrong conclusion too often in the long run (in contrast to the original goal of Neyman and Pearson). Second, the Neyman-Pearson paradigm ignores the proportion of true null hypotheses in the respective field of research. This is problematic as the proportion of true null hypotheses may be quite large and may vary considerably across research fields, due to the Open Science Project and other efforts (e.g., Johnson et al., 2017). As null hypotheses are being tested in an almost industrial scale, it is natural to think of a population of null hypotheses of which a certain proportion is true. When testing a null hypothesis, it is essential to account for the proportion of true null hypotheses in the population from which a

hypothesis originated from to obtain a better understanding of the credibility of the test decision. By ignoring this, there is a high risk of drawing unreliable conclusions and consequently being unable to reproduce (costly) scientific findings.

To counteract these problems, the current paper proposes a shift of the focus from the *classical error rates* which condition on the null and alternative hypotheses being true,

$$\alpha = \mathrm{P}(\text{wrong decision} \mid \mathbf{H_0 \text{ is true}})$$

$$\beta = \mathrm{P}(\text{wrong decision} \mid \mathbf{H_1 \text{ is true}}),$$

towards *conditional error rates* which condition on the decision to reject the null or not, which we denoted by γ and κ:

$$\gamma = \mathrm{P}(\text{wrong decision} \mid \mathbf{\text{reject } H_0})$$

$$\kappa = \mathrm{P}(\text{wrong decision} \mid \mathbf{\text{accept } H_0})$$

to judge the credibility of the outcome of a statistical test. Thus, instead of asking the question "what error rates do we find acceptable conditional on the null and alternative hypothesis being true?", researchers should ask themselves the question "what error rates do we find acceptable when rejecting the null or not?". By conditioning on the known decisions, we expect that researchers have a better understanding of the implied credibility of their research findings when performing a NHST. As these conditional error rates depend on the proportion of true null hypotheses in the field of research, researchers need to take this proportion explicitly into account when working with conditional error rates. Thereby, the classical error rates can be tailored to the research questions at hand (as advocated by Neyman & Pearson) in a more intuitive manner. Also note that others already acknowledged the importance of these alternative conditional error rates (Colquhoun, 2014; 2019; Ioannidis, 2005; Bayarri et al., 2016). Our contribution is a recommendation to use these to check the

credibility of a NHST within the Neyman-Pearson testing paradigm, and to specify the required classical error rates accordingly.

General guidelines were provided for choosing the proportion of true nulls, which includes considerations of broadness of research fields and the viewpoint of target audience within the specified field. To specify the conditional error rates, three guidelines for the selection were provided, such as the choice of so-called changing factors. Furthermore, to stimulate the adoption of the proposed conditional justification for choosing the error rates and to increase awareness, a Shiny app was created for the calculation of the required classical error rates. Researchers are advised to preregister their choices when using conditional justification. We leave the further (practical) implementation of the methodology, including the preregistration of the planned conditional error rates, for future research. Open debates and discussions will be necessary in the scientific community to determine on the desirable values of the conditional error rates in each unique research topic, field, and scenario.

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Appendix A: the Derivation of Equation (5) and (6)

Equation (3) and (4) with $p_1 = 1 - p_0$ lead to the following Equations:

$$\frac{1-\gamma}{\gamma} = \frac{1-\beta}{\alpha} \times \frac{1-p_0}{p_0} \tag{7}$$

$$\frac{1-\kappa}{\kappa} = \frac{1-\alpha}{\beta} \times \frac{p_0}{1-p_0} \tag{8}$$

Solving them for α

$$\alpha = \frac{\gamma(1-p_0)(1-\beta)}{p_0(1-\gamma)} \tag{9}$$

$$\alpha = -\frac{\beta(1-\kappa)(1-p_0) - p_0\kappa}{p_0\kappa} \tag{10}$$

The selection of α depends on p_0, β and γ or κ depending on using Equation (9) or (10). In order to derive the equation for the selection of β. Equation (9) is substituted into Equation (10) and solving it for β, Equation (6) is then obtained for the selection of β

$$\frac{\gamma(1-p_0)(1-\beta)}{p_0(1-\gamma)} = -\frac{\beta(1-\kappa)(1-p_0) - p_0\kappa}{p_0\kappa} \tag{11}$$

$$\beta = \frac{\kappa(\gamma - p_0)}{p_0(1-\kappa-\gamma) + \kappa + \gamma - 1}. \tag{6}$$

By substituting Equation (6) into either Equation (9) or (10) with simplification, Equation (5) is derived

$$\alpha = \frac{\gamma(1-p_0)\left(1 - \frac{\kappa(\gamma - p_0)}{p_0(1-\kappa-\gamma) + \kappa + \gamma - 1}\right)}{p_0(1-\gamma)} \tag{12}$$

$$\alpha = \frac{\gamma(1-p_0)(p_0 - 1 + \kappa)}{p_0(p_0 - \kappa p_0 - \gamma p_0 + \kappa + \gamma - 1)}. \tag{5}$$

Given PPV = 1- γ and NPV = 1- κ, Equation (5) and (6) can be rewritten as a function of PPV and NPV as follows

$$\alpha = \frac{(1-p_0)(p_0 - NPV)(1-PPV)}{p_0(NPVp_0 + p_0PPV - p_0 + 1 - NPV - PPV)} \tag{13}$$

$$\beta = -\frac{NPVp_0 - p_0 - NPV - PPV + NPV\ PPV + 1}{1 - NPV - p_0 - PPV + NPVp_0 + p_0PPV} \tag{14}$$

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Table 1

The probabilities of falsely accepting and rejecting H_0 conditioned on either H_0 or H_1 being true.

	Accepting H_0	Rejecting H_1	Total
H_0 true	True Negative Rate (TNR): $p(Accepting\ H_0 \mid H_0) = 1 - \alpha$	False Positive Rate (FPR): $p(Rejecting\ H_0 \mid H_0) = \alpha$	TNR + FPR = 1
H_1 true	False Negative Rate (FNR): $p(Accepting\ H_0 \mid H_1) = \beta$	True Positive Rate (TPR): $p(Rejecting\ H_0 \mid H_1) = 1 - \beta$	FNR + TPR = 1

Table 2

The probabilities of H_0 or H_1 being true conditional on the testing decision.

	Accepting H_0	Rejecting H_0
H_0 true	Negative Predicting Value (NPV): $p(H_0 \mid Accepting\ H_0) = \frac{p_0(1-a)}{p_0(1-a)+p_1\beta}$	False Discovery Rate (FDR): $\gamma = p(H_0 \mid rejecting\ H_0) = \frac{p_0\alpha}{p_1(1-\beta)+p_0\alpha}$
H_1 true	False Omission Rate (FOR) : $\kappa = p(H_1 \mid Accepting\ H_0) = \frac{p_1\beta}{p_0(1-a)+p_1\beta}$	Positive Predicting Value (PPV): $p(H_1 \mid rejecting\ H_0) = \frac{p_1(1-\beta)}{p_1(1-\beta)+p_0\alpha}$
Total	NPV + FOR = 1	FDR + PPV = 1

Table 3

The procedure of conditional justification and guidelines.

Step	Procedure	Guidelines
(1)	Specify the proportion of true null hypotheses in a research field or research topic at hand.	Two Factors should be considered: • the broadness of the research field • the general viewpoint of the target audience Specific methods: • Results from reproducibility projects and/or meta-analysis. • Experts elicitation. • Equal proportions.
(2)	Specify the desired conditional error rates via the γ and κ levels or their complements (PPV and NPV).	• $\gamma < 0.5$ *and* $\kappa < 0.5$ • $\gamma = \kappa$ • $\gamma < p_0$ and $\kappa < p_1$ by a sufficiently large changing factor • Equations for fulfilling the three criterions above: $\gamma = \kappa = \frac{1}{19 \times \frac{p_1}{p_0} + 1}$ when $p_1 \geq .5$ $\gamma = \kappa = \frac{1}{19 \times \frac{p_0}{p_1} + 1}$ when $p_0 > .5$.
(3)	Get the required classical error rates.	$\alpha = \frac{\gamma(1-p_0)(p_0 - 1 + \kappa)}{p_0(p_0 - \kappa p_0 - \gamma p_0 + \kappa + \gamma - 1)}$ $\beta = \frac{\kappa(\gamma - p_0)}{p_0(1 - \kappa - \gamma) + \kappa + \gamma - 1}$

Table 4

The reproducibility rates across different reproducibility projects in different (sub-) fields.

Project and/or field	Sub-field	Reproducibility rates (and frequency)
The Reproducibility Project: Psychology (Open Science Collaboration, 2015)		36% (35/97)
	Social Psychology	25% (14/55)
	Cognitive psychology	50% (21/42)
Experimental Philosophy (Cova, et al., 2021)		70% (24/34)
Experimental Economics Replication Project (Camerer, et al., 2016)		61% (11/18)
Social Sciences Replication Project (Camerer, et al., 2018)		62% (13/21)

Table 5

The suggested classical error rates for various changing factors and proportions of true null hypotheses.

p_0	Changing factor	PPV/NPV	γ/κ	α	β
25%	19	.9827586	.0172414	.052339901	.005541872
	99	.9966443	.0033557	.01008979	.00111101
	199	.9983278	.0016722	.0050223340	.0005555431
50%	19	.95	.05	.05	.05
	99	.99	.01	.01	.01
	199	.995	.005	.005	.005
75%	19	.9827586	.0172414	.005541872	.052339901
	99	.9966443	.0033557	.00111101	.01008979
	199	.9983278	.0016722	.0005555431	.0050223340

Figure 1

Illustration of a population with a proportion p_0 of true null hypotheses and a proportion p_1 of true alternative hypotheses. Classical error probabilities α and β quantify the rate of drawing the wrong conclusion by conditioning on the unknown true state of the hypotheses (and thus allowing us to ignore the proportions p_0 and p_1), as displayed with the arrow on the right.

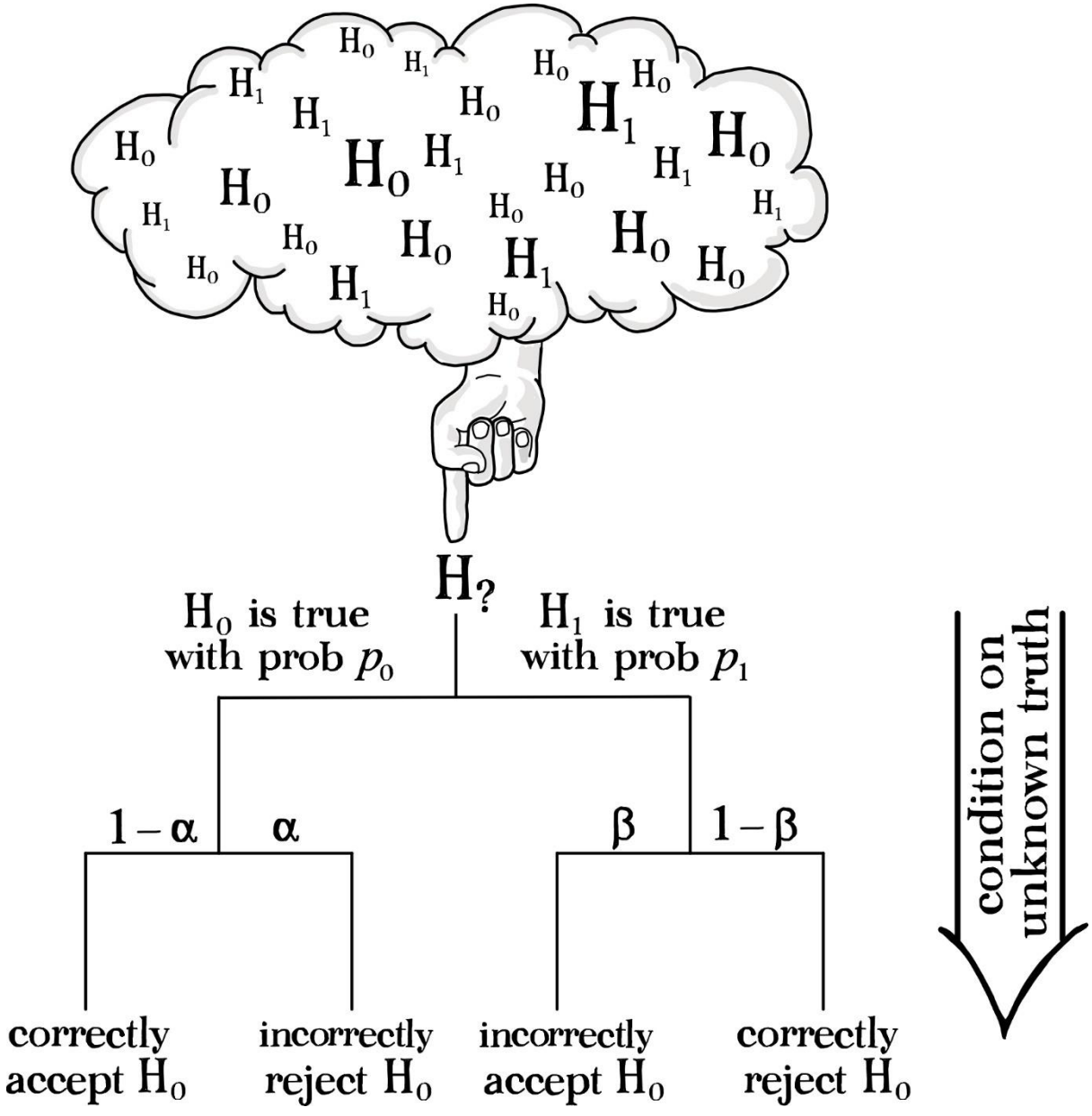


Figure 2

Illustration of alternative condition error probabilities γ and κ which quantify the rate of drawing the wrong conclusion by conditioning on the known decision, as displayed with the arrow on the right.

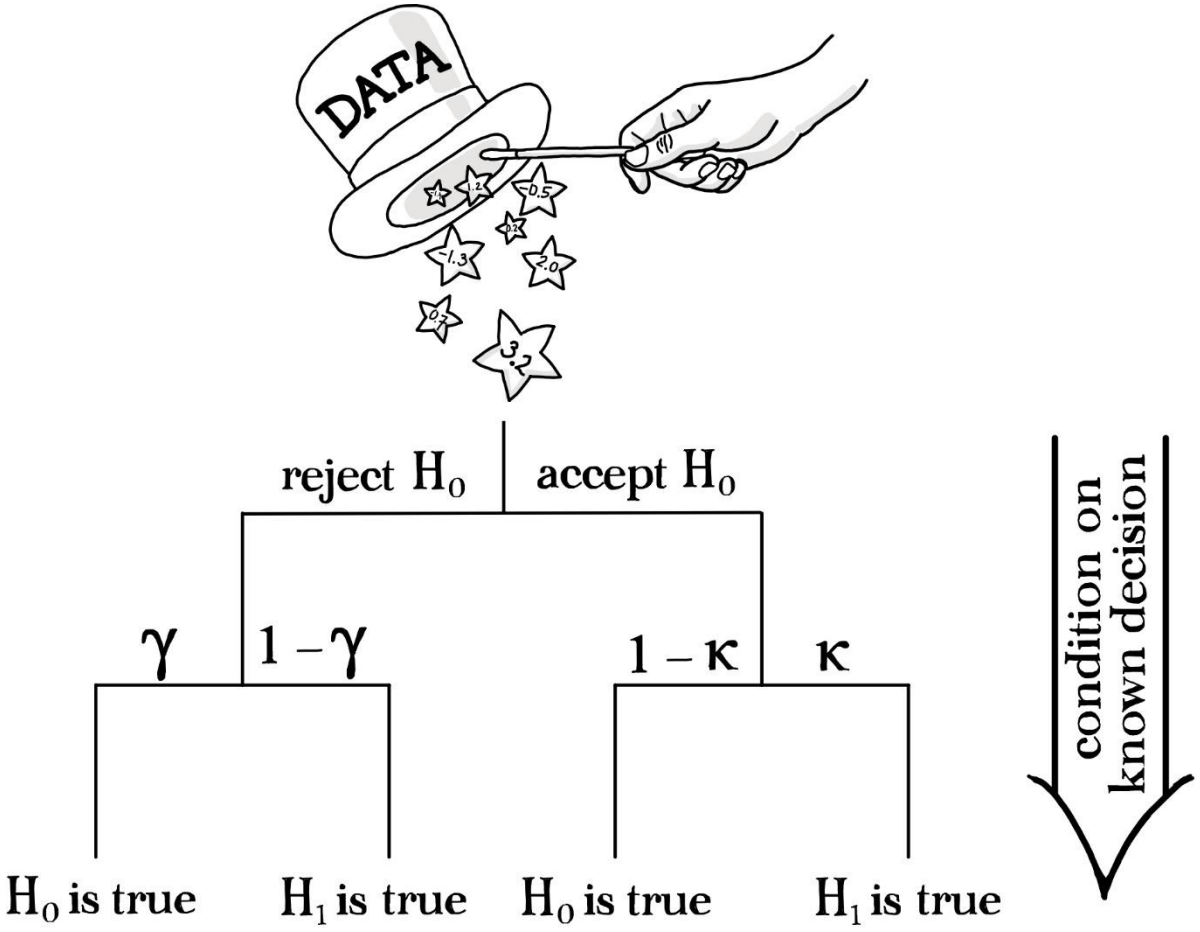
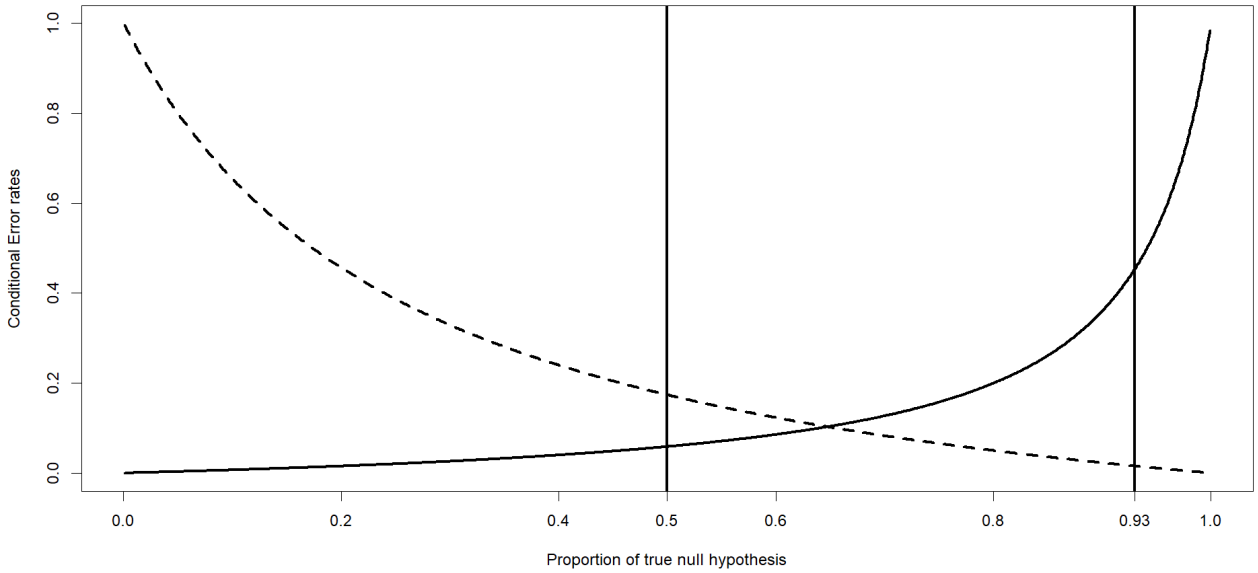


Figure 3

The relationship between conditional error rates and proportion of true null hypotheses.



Note: Solid line indicate the probability of false conclusion given rejection (γ) whereas dashed line represents the probability of false conclusion given acceptance (κ) given α = .05 and β = .2. The vertical lines indicates the proportion = .5 and the estimated proportion of true nulls .93 by Johnson et al. (2017).

Figure 4

The use of the Shiny app

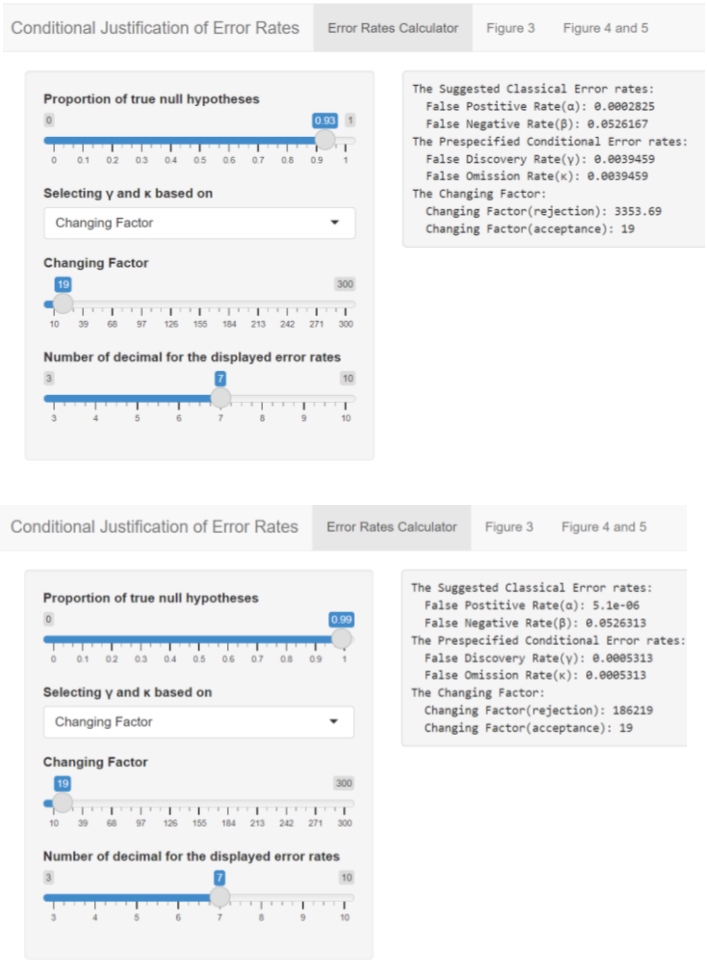
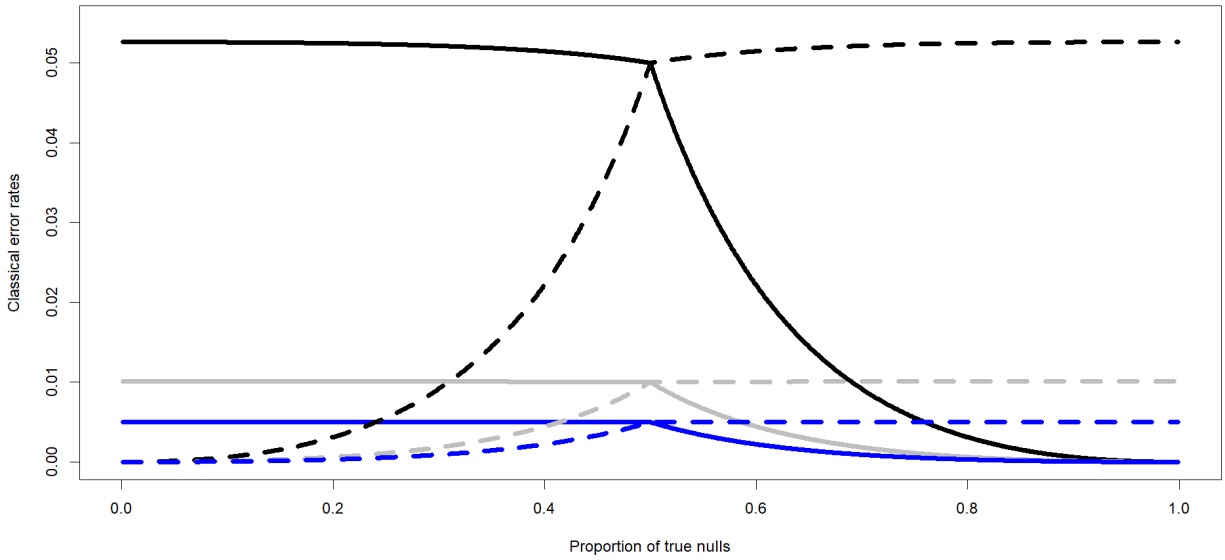


Figure 5

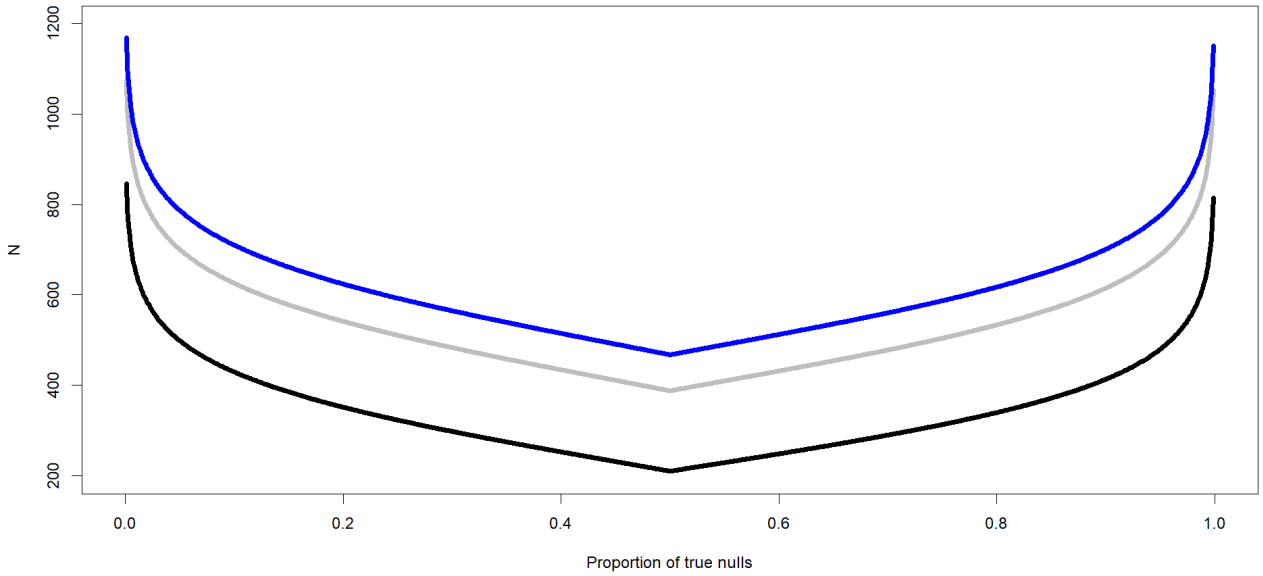
The relationship between proportion of true nulls and required error rates given a set of changing factors.



Note: The solid lines represent the required α based on Equation (5), while the dashed lines depict the required β based on Equation (6). The colors—black, gray, and blue—illustrate the relationship between proportion of true nulls and classcial error rates, considering changing factors of 19, 99, and 199, respectively.

Figure 6

The relationship between the proportion of true nulls and the required sample size given a set of changing factors.



Note: Continuing from the note for Figure 4, the effect size *d* is assumed to be .25 for a two-sided one sample t-test.